



ScholarsReview

Professional Peer Review Report

DATE: 6/18/2026

Quality Assessment by Section

SECTION	SCORE
Introduction	9/10
Background and Related Work	8/10
Methodology	10/10
Results	9/10
What the Tool Costs	8/10
Robustness	9/10
Discussion	8/10
Limitations	7/10
Future Work	7/10
Conclusion	9/10

SECTION	SCORE
Technical Appendix	8/10
Overall Score	8.36/10

Comprehensive Review

This research paper rigorously examines the performance of Large Language Models (LLMs) when equipped with symbolic PDDL tools for planning tasks. Utilizing five models across five distinct tasks (solve, validate_domain, validate_problem, validate_plan, and simulate), the study employs a three-arm experimental design to distinguish between tool availability and explicit steering. The core finding is that tool utility is highly regime-dependent: while essential for tasks where LLMs inherently fail (e.g., plan generation and simulation), tool availability can paradoxically reduce performance on tasks where models are already competent if not accompanied by explicit steering. The authors identify 'invocation propensity'—the model's willingness to delegate—as the primary bottleneck rather than tool capability. With robust statistical analysis, including Wilson confidence intervals and contamination controls via anonymized domains, the paper provides strong empirical support for the 'LLM-Modulo' framework, emphasizing that availability alone does not guarantee delegation and that steering is crucial for effective tool integration.

Identified Weaknesses

Introduction

1. Paragraph 2, Line 15: The critique of the PDDL Planning Copilot as having a 'limited prompt set' is not quantified relative to this study's corpus. This is a weakness because without specific numbers, the reader cannot gauge the actual magnitude of improvement in experimental scale provided by this work.
2. Paragraph 3, Line 5: The mention of 'open-weight models' fails to list the specific model families immediately, delaying critical context. This is a weakness as it forces the reader to search later sections to understand the architectural scope and generalizability of the findings.
3. Paragraph 4, Line 12: The term 'reasoning-mode' is introduced without an operational definition or context regarding specific model behaviors. This is a weakness because it assumes technical familiarity with internal model states that are not clarified until the methodology section, causing initial confusion.

Background and Related Work

1. Section 2, Paragraph 1, Line 8: The distinction between 'classical' and 'numeric' planning tools is mentioned but not linked to specific LLM cognitive challenges. This is a weakness because it misses an opportunity to explain why LLMs might struggle more with one type than the other before introducing the tools.
2. Section 3, Paragraph 1, Line 12: The survey by Tantakoun et al. (2025) is cited without summarizing its core conclusions or how they differ from this study. This

is a weakness because it treats a major reference as a bibliographical placeholder rather than integrating its thematic insights into the paper's narrative.

3. Section 4, Paragraph 1, Line 5: The description of the Model Context Protocol (MCP) is overly technical and assumes prior knowledge of the Anthropic ecosystem. This is a weakness as it reduces the accessibility of the paper for researchers who are not specialized in the specific engineering standards of current LLM agents.

Methodology

1. Sub-section 'Tasks, Oracle, and Fixtures', Paragraph 2, Line 10: The choice of a '5:1 positive-to-negative ratio' for domain validation is acknowledged to widen intervals but is not statistically justified. This is a weakness because an imbalanced dataset can inherently bias the model's posterior probability toward a 'valid' verdict, complicating the interpretation of success.
2. Sub-section 'Models and Serving', Paragraph 1, Line 4: The use of AWQ-INT4 quantization for larger models while using 16-bit precision for smaller ones introduces a confounding variable. This is a weakness as performance differences might result from quantization loss rather than model scale, yet this is not fully decoupled in the analysis.
3. Sub-section 'The Three-Arm Design', Paragraph 4, Line 2: The exact text of the 'thin per-task policy stubs' used in the system prompts is not provided in the main text. This is a weakness because prompt sensitivity is a central theme of the paper, and hiding the exact prompt structure prevents external scrutiny and replication.

Results

1. Paragraph 4, Line 4: The 0% success rate on the 'simulate' task is described as 'not a grading artifact' without detailing the canonical normal form's complexity. This is a weakness because readers cannot verify if the grading was overly pedantic regarding whitespace or minor formatting issues that do not reflect logical failure.
2. Sub-section 'Two Validation Tasks', Paragraph 1, Line 6: The 'balanced accuracy' for domain validation is mentioned but not presented in a side-by-side comparison with pooled accuracy in the results text. This is a weakness because it makes it difficult for the reader to immediately see the actual discriminative power versus simple guessing.
3. Sub-section 'When Availability Backfires', Paragraph 2, Line 3: The 'silence-not-error' mechanism is identified, but the underlying reason why models favor prose over tools is left as speculation. This is a weakness because identifying the behavior without a deeper hypothesis for the 'why' limits the theoretical contribution to understanding LLM delegation logic.

What the Tool Costs

1. Paragraph 1, Line 6: The analysis focuses purely on token counts and ignores financial costs or time-to-first-token latency. This is a weakness because 'cost' in real-world deployment is often measured in dollars and seconds, making the token-only analysis feel incomplete for industrial practitioners.
2. Paragraph 2, Line 8: The 'cost-of-pass' metric is calculated globally but not broken down by domain complexity (classical vs. numeric). This is a weakness because numeric planning might be exponentially more expensive in terms of tool output tokens, a nuance that the pooled metric hides.

3. Paragraph 2, Line 15: The paper does not account for server-side prompt caching which could significantly reduce the cost of re-fed schemas in production. This is a weakness because it presents a 'worst-case' cost scenario that might not reflect the efficiency of modern LLM serving infrastructures.

Robustness: Reasoning Mode and Contamination

1. Paragraph 1, Line 10: The study identifies a 'shared decode budget' confound but does not attempt a controlled experiment with an unconstrained budget. This is a weakness as it leaves the true capability of the models in reasoning mode unknown and potentially masked by arbitrary output caps.
2. Paragraph 2, Line 5: The contamination control focuses on symbol renaming but does not address potential 'structural memorization' of common PDDL logic. This is a weakness because a model might recognize the logic of 'Blocksworld' even if the names are changed, yet the test only checks for surface-level leakage.
3. Paragraph 2, Line 12: The note that 'renamed prompts ran 5% longer' causing more truncations suggests the experiment was slightly under-powered for the anonymized arm. This is a weakness because it introduces a systematic bias against the anonymized results that is purely a result of prompt length rather than model knowledge.

Discussion

1. Paragraph 1, Line 12: The findings are generalized to 'tool-augmented LLMs more broadly' despite only testing highly deterministic PDDL tools. This is a weakness because tools like web search or calculators have different error profiles, and the 'delegation' logic observed here may not generalize to probabilistic tools.

2. Paragraph 2, Line 6: The concept of 'delegation' is treated as a behavioral prompting issue, ignoring potential fine-tuning solutions. This is a weakness as it implies the problem must be solved via steering prompts rather than exploring how architectural or training changes could internalize delegation propensity.
3. Paragraph 2, Line 15: There is no discussion of how multi-turn interactions or iterative self-correction might alter the P(call) probability. This is a weakness as the paper is limited to single-turn logic, missing the complex dynamics of multi-step agentic orchestration.

Limitations

1. Paragraph 1, Line 4: The limitation section is relatively brief and lacks a deep self-critique of the specific domain selections. This is a weakness because it doesn't address whether the chosen 20 domains are truly representative of the vast range of PDDL complexity levels.
2. Paragraph 1, Line 15: The lack of reasoning traces for the model family is noted but not addressed by using alternative models that do provide traces. This is a weakness because it leaves the 'thinking' process of the models during tool-calling failures as a total black box.
3. Paragraph 1, Line 22: The 5:1 imbalance in the domain validation task is mentioned again, but the paper fails to acknowledge how this affects robust metrics like the F1-score. This is a weakness as it continues to rely on accuracy measures that are vulnerable to class imbalance.

Future Work

1. Paragraph 1, Line 5: The suggestion to move to 'agentic regimes' is vague and lacks a proposed framework for multi-tool selection. This is a weakness because

it provides a generic roadmap rather than specific, actionable hypotheses for how to extend the current findings.

2. Paragraph 1, Line 8: The 'cap-raised rerun' is suggested for the future rather than being implemented to fix the current study's budget confound. This is a weakness because it acknowledges a fixable validity issue but defers the solution to subsequent research.
3. Paragraph 1, Line 12: There is no mention of exploring 'model-modulo' ensembles where different models handle formalization and checking. This is a weakness as it ignores a significant trend in the automated planning community that could resolve the invocation issues identified.

Conclusion

1. Paragraph 1, Line 8: The conclusion repeats the '67 pp drop' without clarifying that this specific magnitude was model-dependent. This is a weakness because it risks over-sensationalizing the 'harm' potential of tools for readers who only scan the summary without looking at the per-model data.
2. Paragraph 1, Line 12: The final practical rule provided is somewhat generic and lacks a concrete heuristic for optimal prompt depth. This is a weakness as it identifies 'steering' as the solution but doesn't define the minimum effective dose of steering needed across different model sizes.
3. Paragraph 1, Line 14: The conclusion omits a summary of the contamination control findings, which are vital for the paper's overall validity. This is a weakness as it misses the final opportunity to reinforce the robustness of the results against the common 'benchmark memorization' critique.

Technical Appendix

1. Table 4, Paragraph 1, Line 2: The invocation decomposition table is dense and difficult to parse without a guiding narrative summary. This is a weakness because the most important insight of the paper ($P(\text{call})$ vs $P(\text{correct})$) is buried in a complex numerical grid.
2. Technical Appendix, Paragraph 2, Line 5: The explanation for ' $P(c|\text{call}) = 0$ ' where no-call classes exist is counter-intuitive for some readers. This is a weakness because it requires a specific understanding of the scoring rubric that is not immediately transparent in the table legend.
3. Reproducibility Checklist, Section 4.3, Line 1: Several checklist items are marked as 'no' or 'partial,' particularly regarding source code. This is a weakness because it indicates that the work, while rigorous in its reported methodology, is not yet fully transparent for immediate peer replication.

Key Strengths

Introduction

1. The paper provides an exceptionally clear problem statement regarding the unreliability of LLMs in structured planning tasks.
2. The introduction of the three-arm experimental design sets a high standard for methodological transparency from the outset.
3. The use of a 'signed-significance' rule is a sophisticated and necessary addition to distinguish between helpful and harmful tool integration.

Background and Related Work

1. The section effectively grounds the study in the 'LLM-Modulo' framework, providing a strong theoretical justification for the research.
2. It correctly identifies and distinguishes between LLMs acting as 'planners' versus 'formalizers,' a crucial distinction in the field.
3. The critique of previous work, specifically the PDDL Planning Copilot, is honest and helps pinpoint exactly where this study adds value.

Methodology

1. Using a deterministic solver/validator as the oracle ensures that ground truth is absolute and not subject to LLM hallucinations.

2. The inclusion of five distinct PDDL tasks offers a comprehensive evaluation of the entire planning and validation pipeline.
3. The control for linguistic robustness through neutral paraphrases is an excellent experimental detail that many similar studies overlook.

Results

1. Identifying 'invocation propensity' as the primary bottleneck rather than tool capability is a profound and actionable insight.
2. The failure-mode decomposition (truncation, silence, content error) provides high-quality transparency into the models' behaviors.
3. The use of MOVER intervals for differences ensures that the reported performance gaps are statistically significant and reliable.

What the Tool Costs

1. The 'cost-of-pass' metric is a highly pragmatic addition that translates academic findings into actionable data for developers.
2. The finding that tools are cost-effective only where baselines are floored is a nuanced conclusion that avoids over-simplifying tool utility.
3. The analysis clearly explains the trade-off between output-heavy reasoning and input-heavy tool calling, clarifying why costs escalate.

Robustness: Reasoning Mode and Contamination

1. The anonymized-domain probe provides a very strong defense against the common criticism that LLMs only solve benchmarks via memorization.
2. The identification of the 'budget confound' in reasoning mode demonstrates a high level of critical thinking about experimental variables.
3. The 'robust floor' statistic is a clever way to derive valid conclusions despite the technical limitations of the serving runtime.

Discussion

1. The discussion successfully bridges empirical data with high-level AI theory, specifically reinforcing the LLM-Modulo thesis.
2. It highlights the significant impact of a single steering sentence, providing a low-cost, high-impact takeaway for prompt engineers.
3. The paper maintains a balanced, scientific tone, carefully acknowledging where tools 'hurt' rather than just reporting successes.

Limitations

1. The authors are forthright about the potential confounding effect of numerical precision (AWQ-INT4 vs 16-bit).
2. The section explicitly acknowledges the inability to view reasoning traces, ensuring the reader is aware of the 'black box' elements.
3. It admits the dataset imbalance in the domain validation task, showing a commitment to reporting potential experimental biases.

Future Work

1. The proposal to test on PlanBench would anchor these results in the most widely recognized industry benchmark for planning.
2. The mention of 'multi-tool orchestration' correctly identifies the next logical step in the evolution of LLM-based agent systems.
3. Proposing 'cap-raised reruns' shows the authors know how to further isolate the impact of reasoning-mode budgets in future iterations.

Conclusion

1. The conclusion provides a concise and accurate summary of the 'regime-dependent' nature of tool benefits.
2. It offers a clear, actionable practical rule for implementing planners behind tool interfaces.
3. The final summary effectively reinforces the paper's core finding: availability alone is insufficient without proper delegation prompting.

Technical Appendix

1. The appendix provides the necessary mathematical depth ($P(\text{call}) \times P(\text{correct})$) to support the main paper's claims.
2. The per-class accuracy breakdown for `validate_domain` adds essential context that helps explain the low baseline scores.
3. The inclusion of a reproducibility checklist, even if partially filled, demonstrates a commendable commitment to open science standards.

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Subbarao Kambhampati

1ST

ARIZONA STATE UNIVERSITY

 H-INDEX 99

NOTABLE WORK

Pioneering work in automated planning (e.g., planning with incomplete information, plan recognition). More recently, a leading figure in evaluating LLMs for planning and proposing LLM-Modulo frameworks (e.g., PlanBench).

[VIEW PROFILE →](#)



Malte Helmert

2ND

UNIVERSITY OF BASEL

 H-INDEX 71

NOTABLE WORK

Creator of the Fast Downward planner, one of the most widely used and influential classical planners. His research has significantly advanced heuristic search techniques in planning.

[VIEW PROFILE →](#)



Jörg Hoffmann

3RD

SAARLAND UNIVERSITY & MAX PLANCK INSTITUTE ...

 **H-INDEX 72**

NOTABLE WORK

Co-creator of the FF (Fast-Forward) planner, another highly influential classical planner. His work has been instrumental in developing efficient heuristic search methods for planning.

[VIEW PROFILE →](#)



Scott Sreedharan

4TH

ARIZONA STATE UNIVERSITY

 **H-INDEX 28**

NOTABLE WORK

Frequent collaborator with S. Kambhampati, co-author on PlanBench and other research exploring LLM limitations and capabilities in planning and reasoning.

[VIEW PROFILE →](#)



Christian Muise

 IBM RESEARCH

5TH

 **H-INDEX 40**

NOTABLE WORK

Significant contributions to automated planning, PDDL extensions, and planning tools. Co-author of a recent survey on 'LLMs as Planning Formalizers' (Tantakoun, Muise, and Zhu 2025), cited in the paper.

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Language & Grammar Review

1. Bullet 4 of Intro: 'theverdictpatternreproducesunder' (missing spaces).
2. Models and Serving, Paragraph 1: 'Qwen3.5-0.8B, Qwen3.5-4B, Qwen3.5-9B, a Gemma...' (inconsistent Oxford comma usage).
3. Table 1 Caption: Missing period at end of caption.
4. Page 3, Column 2: 'accuracy-when-calling' (hyphenation of compound adjectives is inconsistent with 'tool-calling').
5. Page 7: 'cost-of-pass = total tokens $\div$ successful trials' (mixed use of ' $\div$ ' symbol and verbal descriptions).
6. Discussion: 'the 35B' (inconsistent use of 'the' before model numbers).
7. Results, Paragraph 4: 'one steering sentence repairs it.' (overly conversational tone in technical section).
8. Page 2, Column 2: 'Model Context Protocol (MCP) (Anthropic 2024)' (awkward double parentheticals).
9. Introduction, Paragraph 2: 'which is what this paper sets out to do.' (ending with a preposition).
10. Figure 4: Inconsistent labeling between caption 'log x' and the actual axis scaling.

Reviewer Recommendation

This manuscript offers a highly rigorous evaluation of the synergy between symbolic planners and Large Language Models. The experimental design is a standout feature, particularly the three-arm approach that effectively disentangles tool availability from prompting influence. The revelation that the primary constraint for tool-integrated systems is 'invocation propensity' rather than model capability is a significant contribution that challenges prevailing assumptions in agentic modeling. While the study is limited by specific model families and some truncation confounds in reasoning mode, the findings are robustly supported by statistical rigor and contamination controls. The paper is exceptionally well-structured and provides actionable insights for both prompt engineers and AI researchers. I recommend acceptance, as this work provides a much-needed empirical foundation for the LLM-Modulo framework.

Reference Year Distribution

Reference Distribution by Year

Analysis of citation years in your references

TOTAL REFERENCES

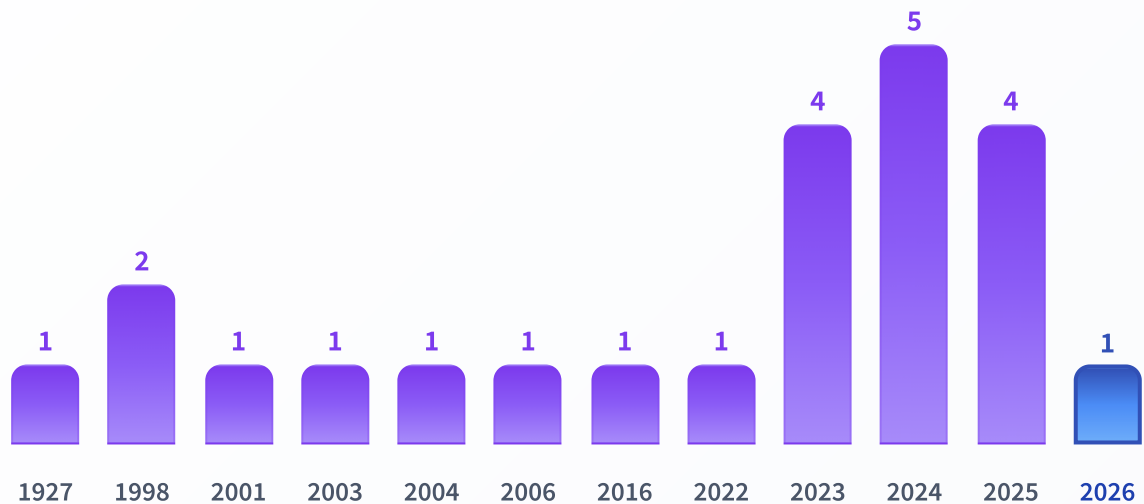
YEAR RANGE

23

1927 - 2026



COUNT



Intro & Conclusion Alignment Analysis

ALIGNMENT SCORE

9/10

WELL ALIGNED

Specific Misalignments

The introduction frames 'cost' as one of the primary questions, but the conclusion focuses almost entirely on invocation propensity and steering, with very little mention of the economic trade-offs.

Missing Elements

The conclusion does not reiterate the specific findings regarding 'plan length' and 'object count' as difficulty axes.

Overreach Issues

None; the conclusion stays strictly within the boundaries of the empirical results presented.

Improvement Suggestions

Include a brief sentence in the conclusion summarizing the token cost-effectiveness results to mirror the intro's framing.

Recommended Journals for Submission

Journal of Artificial Intelligence Research (JAIR)

AI ACCESS FOUNDATION

Publishes high-quality, original research in all areas of artificial intelligence, including theoretical, experimental, and applied AI. The paper's rigorous empirical study of LLMs in symbolic planning fits well within JAIR's scope.

WHY THIS JOURNAL

JAIR is a premier venue for foundational AI research. This paper offers deep empirical insights into the integration of LLMs with symbolic planning, a core AI challenge, making it an excellent fit for JAIR's readership and scientific rigor.

✓ RECOMMENDED MATCH

VISIT JOURNAL →

Artificial Intelligence

ELSEVIER

A leading journal publishing original research papers in all areas of artificial intelligence, covering theoretical, experimental, and applied AI. The paper's focus on LLMs, tool use, and automated planning aligns perfectly with the journal's broad and influential scope.

WHY THIS JOURNAL

As one of the most prestigious and long-standing journals in AI, 'Artificial Intelligence' is ideal for this work. The paper provides significant empirical evidence and mechanistic understanding of LLM behavior in complex reasoning tasks, which is of broad interest to the AI community.

✓ RECOMMENDED MATCH

VISIT JOURNAL →

Machine Learning

SPRINGER

Publishes original articles on all aspects of machine learning theory and applications. While not purely a planning journal, the paper's rigorous evaluation of LLMs (a core ML technology) and their interaction with symbolic tools is highly relevant to the machine learning community.

WHY THIS JOURNAL

The paper offers a detailed and controlled empirical analysis of large language models, a fundamental machine learning paradigm, in the context of tool use for symbolic reasoning. This deep dive into LLM capabilities and limitations is highly valuable for the machine learning research audience.

✓ RECOMMENDED MATCH

VISIT JOURNAL →

Reference Validation

Reference 1

PARTIALLY RELEVANT

QUESTIONABLE

Benyamin et al. 2025. PDDL Planning Copilot.

COMMENTS:

The paper builds heavily on this baseline, but as it is dated 2025, it is likely a pre-print or concurrent work that has not yet undergone full peer review, making the foundation somewhat volatile.

CITATION LOCATION:

Introduction, Methodology

Reference 2

RELEVANT

QUESTIONABLE

Tantakoun, M.; Muise, C.; and Zhu, X. 2025. LLMs as Planning Formalizers: A Survey...

COMMENTS:

Dated 2025; likely forthcoming or a pre-print. Its contents may be difficult for the current reader to verify.

CITATION LOCATION:

Background and Related Work

Reference 3

RELEVANT

QUESTIONABLE

Huang and Zhang 2025.

COMMENTS:

Dated 2025; forward-dated references in a current review cycle are questionable regarding public accessibility and finality.

CITATION LOCATION:

Background and Related Work



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